Laboratory 01

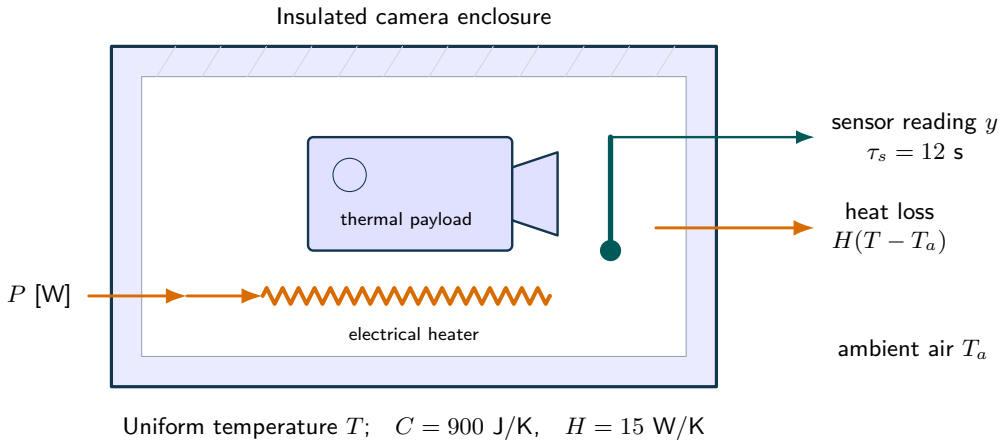
Thermal enclosure and a slow sensor

Exercise sheet – five integrated problems

Evidence boundary. Original hypothetical scenarios. Derive, calculate and justify decisions; every numerical record is synthetic. Figures are conceptual and not to scale. **Shared nominal data.** Nominal enclosure: $C = 900$ J/K, $H = 15$ W/K, sensor $\tau_s = 12$ s. Input P is heat rate; x is physical temperature rise and y is the indicated rise. A 40 W step starts from ambient unless a part states another condition.

1. A thermal enclosure under competing design requirements

An insulated instrument box must warm its payload while losing heat through its walls. Use the physical view to identify stored energy, heater input and ambient heat leakage.



Cutaway of a lumped thermal enclosure. Orange arrows carry heat; the probe measures a lagged temperature. The drawing fixes the energy boundary, not a real device geometry.

(a) An enclosure has thermal capacitance $C = 900$ J/K and conductance $H = 15$ W/K. With constant ambient temperature, define $x = T - T_a$ and derive the balance from heat input P to x. Give the units of every term and the zero-state transfer function.

(b) Keep $C = 900$ J/K but increase H from 15 to 20 W/K. Compare the time constants and the steady rises for 40 W. Find the heater power required to maintain a 3 K rise in each design.

(c) A redesigned box must maintain a 3 K rise with at most 45 W and have thermal time constant no greater than 45 s. Keep $C = 900$ J/K and vary only H. Derive both constraints and decide whether any design is feasible.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

2. A reassuring sensor reading can hide stored heat

The nominal box receives a 40 W step from ambient. Its temperature sensor has its own state and a 12 s lag. Evaluate the largest disagreement between the payload and its indicated temperature.

(a) A sensor obeys $12\dot{y} = x - y$, initially $y = 0$. For the 40 W heat step, derive Y/P, $y(t)$ and both initial slopes. Explain the different relative degrees.

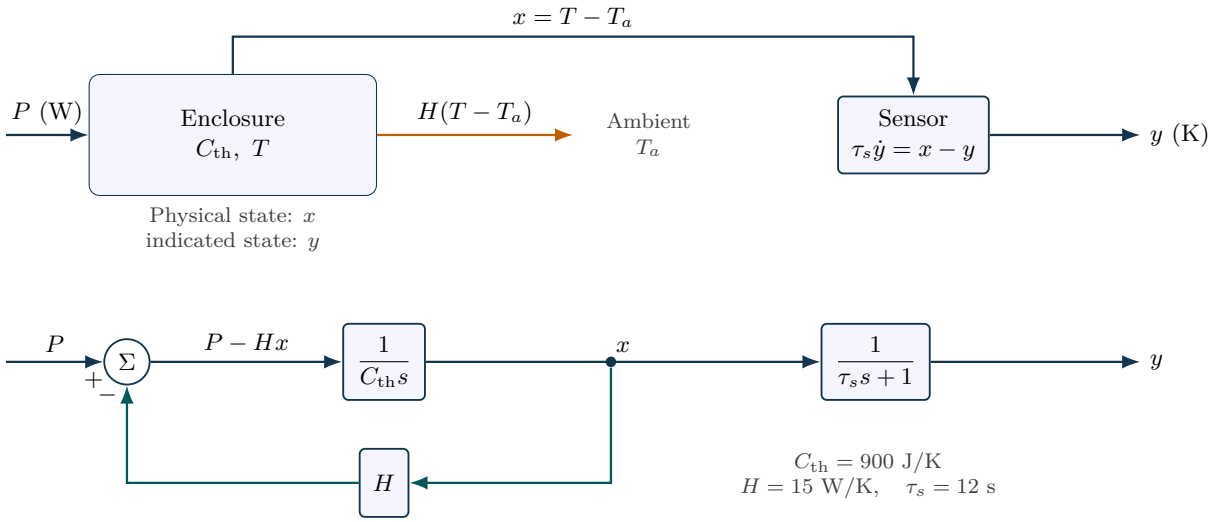
(b) For the previous step, find the time and magnitude of the largest positive error $e = x - y$. Verify the error is zero initially and vanishes as time tends to infinity.

(c) Suppose a limit is checked using only the indicated rise y. Find the smallest constant allowance Δ such that $x(t) \leq y(t) + \Delta$ for this particular step. Explain why that allowance is not a universal safety margin.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

3. A heating pulse with a finite energy budget

The heater can operate at 40 W for only one minute. A 2 K target is desired, followed by passive cooling. Reconcile the continuous-heating prediction with the pulse experiment.



Thermal storage, heat leakage and sensor dynamics are separate elements. The lower loop implements the energy balance, not an added temperature controller.

(a) Starting at ambient, apply $P = 40$ W. Find the equilibrium rise, initial slope and time to reach a rise of 2 K. Is a 3 K target reachable with this constant input?

(b) Apply 40 W for 60 s and then switch the heater off. Derive the physical temperature rise before and after switch-off, and determine the additional cooling time needed to fall to 0.5 K.

(c) Compute supplied energy, stored incremental energy and heat lost by the switch-off time. Use the balance to explain why the pulse cannot reach the 2 K target even though continuous heating can.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

4. Identifying a thermal process through an imperfect instrument

A synthetic characterization record is available, but the instrument may have unmodeled lag. Assess which parameter conclusions survive that uncertainty and whether inverse filtering fixes the problem.

(a) A deliberately synthetic test gives a 3 K equilibrium rise for a 36 W step. A direct physical-state measurement reaches $1 - e^{-1}$ of its final value after 75 s. Identify H and C. Explain what changes if the dynamic observation comes from an unmodeled slow sensor.

(b) An idealized inverse sensor uses $\hat{x}_k = y_k + 12(y_k - y_{k-1})/\Delta t$. For $\Delta t = 1$ s and independent sensor errors of standard deviation 0.02 K, calculate the standard deviation in $\hat{x}$. Explain the consequence of reducing the sampling interval.

(c) Design a two-stage identification and validation protocol that distinguishes heat leakage from storage and sensor lag. State which conclusions cannot be obtained from the quoted equilibrium and fraction-crossing observations alone.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.

5. A warm start during an ambient-temperature change

Warm initial conditions and external heating act through different terms of the same balance. Keep the distinction between absolute temperature deviation and rise above the current ambient.

(a) Use absolute temperatures and a 12 s sensor. Derive the transfers from heater power and ambient-temperature deviation to measured temperature. Evaluate the final measured change caused by a 30 W power increment and a simultaneous 2 K ambient increase.

(b) At switch-on the enclosure rise is 1 K, the sensor indicates 0 K, and power becomes 40 W. Derive $x(t)$ and $y(t)$ for the nominal parameters. State why multiplying a transfer function by $40/s$ alone is insufficient.

(c) Combine the two effects: use absolute deviations with $\Delta T(0) = 1$ K, $y(0) = 0$, a 30 W heater increment and a simultaneous 2 K ambient increment. Derive $x(t) = \Delta T(t)$ and the measured deviation $y(t)$, and check initial and final values.

Submission. Show the model, intermediate reasoning, units and a conclusion addressing the stated requirements.